Math 661 Applied Statistics – Module 12 Assignment

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Task 1 – How did you use your cell phone?

A Pew Internet poll surveyed 1003 adults living in the United States by telephone. Of these, 462 responded that during the past 30 days they had used their phone while in a store to call a friend or family member for advice about a purchase they were considering.

Part 1

(a) Identify the sample size and the count

The sample size is the total number of individuals included in the survey. The count is the number of sampled individuals who gave the response of interest (here, "yes").

Sample size: $n = 1003$

Count of "yes" responses: $x = 462$

Therefore, the sample consists of 1003 adults, and 462 of those adults said they had used their phone in a store in this way during the previous 30 days.

(b) Calculate the sample proportion

The sample proportion is obtained by dividing the count by the sample size:

$\hat{p} = x / n$

$\hat{p} = 462 / 1003$

$\hat{p} = 0.460618$

$\hat{p} \approx 0.4606 = 46.06\%$

So, about 46.06% of the adults in the sample answered "yes."

(c) Explain the relationship between the population proportion and the sample proportion

The population proportion, denoted by p, is the true proportion of all U.S. adult cell phone owners who would answer "yes" to this question. In practice, that true population value is unknown, because it is usually impossible to ask every individual in the population.

The sample proportion, denoted by $\hat{p}$, is computed from the sample and serves as an estimate of p. Thus, $\hat{p}$ is used to learn about p. Because $\hat{p}$ depends on which sample happened to be selected, it is subject to sampling variability. However, if the sample is reasonably representative and large enough, $\hat{p}$ should be close to p.

Part 2

(a) Report the sample proportion, the standard error of the sample proportion, and the margin of error for 95% confidence

From Part 1, the sample proportion is:

$\hat{p} = 0.460618 \approx 0.4606$

The standard error measures how much the sample proportion is expected to vary from sample to sample when random sampling is used. The formula is:

$SE(\hat{p}) = \sqrt{[\hat{p}(1 - \hat{p})/n]}$

Substitute the observed values:

$SE(\hat{p}) = \sqrt{[(0.460618)(0.539382) / 1003]}$

$SE(\hat{p}) = \sqrt{[0.248449 / 1003]}$

$SE(\hat{p}) = \sqrt{[0.000247706]}$

$SE(\hat{p}) = 0.015739 \approx 0.0157$

For a 95% confidence interval, the critical value is $z^* = 1.96$. The margin of error is therefore:

$ME = z^* \times SE(\hat{p})$

$ME = 1.96 \times 0.015739$

$ME = 0.030848 \approx 0.0308$

Hence the required summary quantities are: sample proportion 0.4606, standard error 0.0157, and margin of error 0.0308.

(b) Are the guidelines for when to use the large-sample confidence interval for a population proportion satisfied in this setting? Explain your answer.

Yes, the conditions are satisfied. The large-sample confidence interval for a population proportion requires that the sample be reasonably representative, that the population is much larger than the sample (the 10% condition), and that the numbers of successes and failures in the sample are both at least 10.

Check the large-counts condition explicitly:

$n\hat{p} = 1003 \times 0.460618 = 462$

$n(1 - \hat{p}) = 1003 \times 0.539382 = 541$

Both values are far greater than 10, so the Normal approximation is appropriate. In addition, 1003 is much less than 10% of the number of U.S. adult cell phone owners, so the independence condition is also reasonable.

(c) Find the 95% large-sample confidence interval for the population proportion

The generic form of the confidence interval is:

$\hat{p} \pm ME$

Substitute the values already found:

0.460618 ± 0.030848

Now calculate the endpoints one at a time:

Lower bound = 0.460618 − 0.030848 = 0.429770

Upper bound = 0.460618 + 0.030848 = 0.491466

Thus, the 95% large-sample confidence interval for the population proportion is (0.4298, 0.4915).

In percentage form, the interval is approximately 42.98% to 49.15%.

(d) Write a short statement explaining the meaning of your confidence interval

We are 95% confident that the true proportion of U.S. adult cell phone owners who used their phone while in a store during the past 30 days to call a friend or family member for advice about a purchase is between 42.98% and 49.15%.

This interpretation refers to the unknown population proportion. It does not mean that 95% of individual people are in the interval; rather, it means that the interval procedure used here captures the true population proportion in about 95% of repeated samples of this kind.

Quantity	Symbol	Value	Comment
Sample size	n	1003	Number of adults surveyed
Count	x	462	Number answering “yes”
Sample proportion	$\hat{p}$	0.4606	Observed proportion of “yes” responses
Standard error	$SE(\hat{p})$	0.0157	Sampling variability of $\hat{p}$
Margin of error (95%)	ME	0.0308	Half-width of the confidence interval
95% CI		(0.4298, 0.4915)	Plausible range for the population proportion

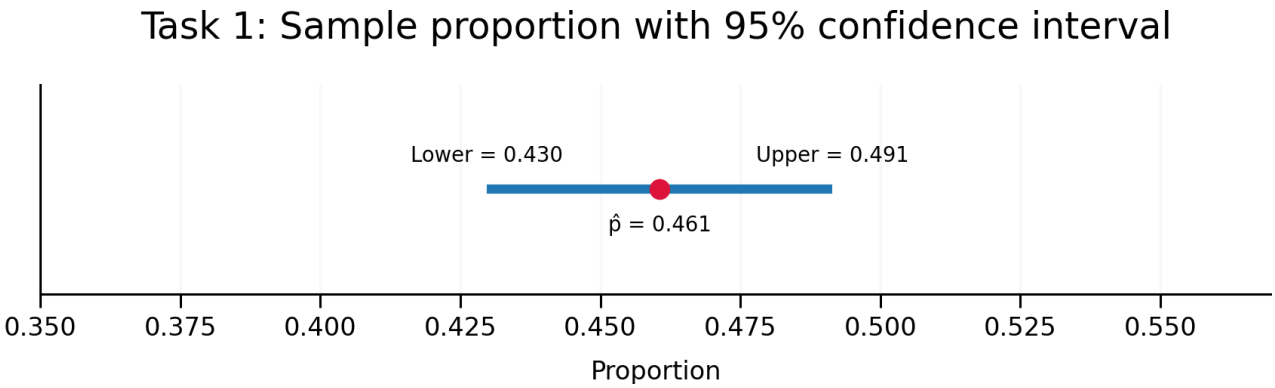

Figure 1. Task 1 sample proportion with the 95% confidence interval shown on a number line.

Task 2 – Exergaming in Canada (IPS Exercise 8.42)

Exergames are active video games such as rhythmic dancing games, virtual bicycles, balance board simulators, and virtual sports simulators that require a screen and a console. A study of exergaming among Grade 10 and Grade 11 students in Montreal, Canada, examined factors related to participation in exergaming. Of the 358 students who reported that they stressed about their health, 29.9% said that they were exergamers. Of the 851 students who reported that they did not stress about their health, 20.8% said that they were exergamers.

(a) Define the two populations to be compared for this exercise

The first population is all Grade 10 and Grade 11 students in Montreal who stress about their health. The second population is all Grade 10 and Grade 11 students in Montreal who do not stress about their health. The goal is to compare the population proportion of exergamers in these two populations.

(b) What are the counts, sample sizes, and the proportions?

The problem gives the sample sizes and the sample proportions directly, but not the counts. Because the percentages are rounded to one decimal place, the counts have to be inferred and are therefore approximate.

For the group that stresses about health:

$n_1 = 358$

$\hat{p}_1 = 0.299$

$x_1 \approx n_1\hat{p}_1 = 358 \times 0.299 = 107.042 \approx 107$

Failures in group 1 $\approx 358 - 107 = 251$

For the group that does not stress about health:

$n_2 = 851$

$\hat{p}_2 = 0.208$

$x_2 \approx n_2\hat{p}_2 = 851 \times 0.208 = 177.008 \approx 177$

Failures in group 2 ≈ 851 − 177 = 674

So the "stress" group has about 107 exergamers out of 358 students, and the "no-stress" group has about 177 exergamers out of 851 students.

Group	Count of exergamers (x)	Sample size (n)	Sample proportion ($\hat{p}$)
Stress about health	107	358	0.299
Do not stress about health	177	851	0.208

Because the proportions 29.9% and 20.8% were given as rounded percentages, the exact counts are not guaranteed to be integers before rounding. The values 107 and 177 are the most reasonable inferred counts.

(c) Are the guidelines for the use of the large-sample confidence interval satisfied?

For a two-sample confidence interval for a difference in population proportions, the main conditions are: independent groups, a sufficiently small sample relative to the population, and at least 10 successes and 10 failures in each group.

Check the success–failure condition explicitly:

Group 1 successes ≈ 107, failures ≈ 251

Group 2 successes ≈ 177, failures ≈ 674

All four numbers are larger than 10. The two groups are also naturally separate (students who stress about health versus students who do not), so treating the groups as independent is reasonable. Therefore, the large-sample confidence interval conditions are satisfied.

(d) Are the guidelines for the use of the large-sample significance test satisfied?

For a two-sample significance test comparing proportions, the expected numbers of successes and failures are checked using the pooled sample proportion.

$\hat{p}_c = (x_1 + x_2) / (n_1 + n_2)$

$\hat{p}_c = (107 + 177) / (358 + 851)$

$\hat{p}_c = 0.234905 \approx 0.235$

Now compute the expected counts in each group under the pooled estimate:

Group 1 expected successes = $n_1\hat{p}_c = 358 \times 0.234905 = 84.096$

Group 1 expected failures = $n_1(1 - \hat{p}_c) = 358 \times 0.765095 = 273.904$

Group 2 expected successes = $n_2\hat{p}_c = 851 \times 0.234905 = 199.904$

Group 2 expected failures = $n_2(1 - \hat{p}_c) = 851 \times 0.765095 = 651.096$

All expected counts are greater than 10, so the large-sample significance test conditions are also satisfied.

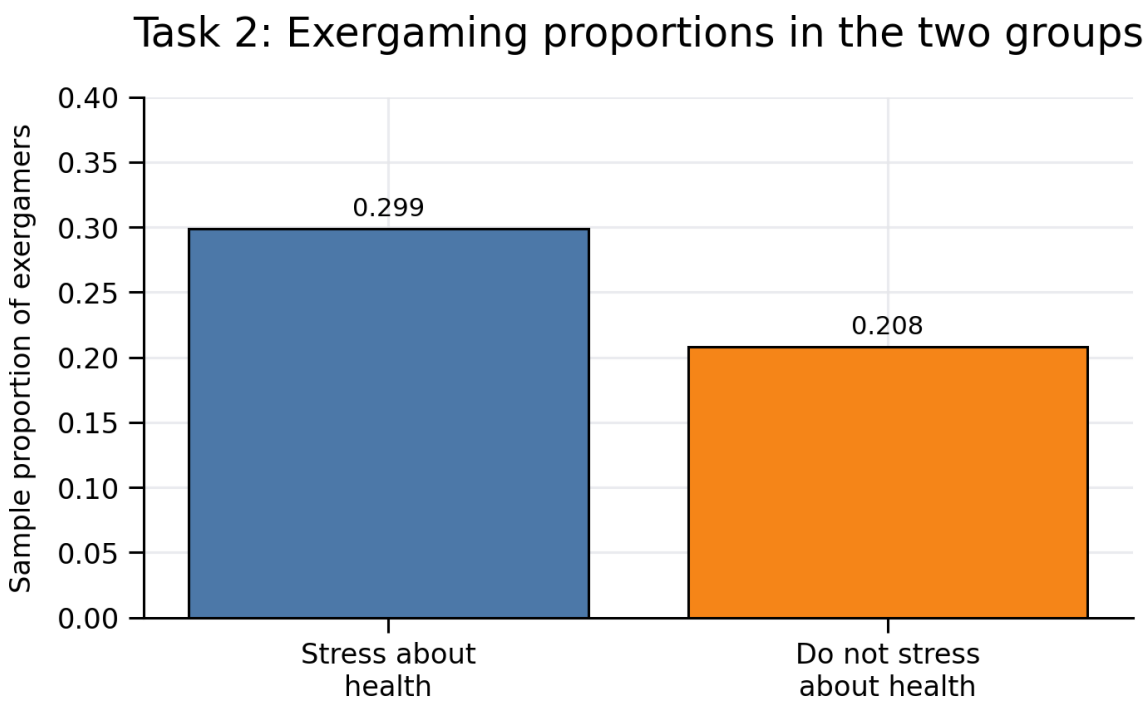

Figure 2. Task 2 comparison of the sample proportions of exergamers in the two student groups.

Task 3 – Some probability distributions (IPS Exercise 4.115)

The random variable X has the following probability distribution: X can take the values 4, 5, and 6 with probabilities 0.3, 0.4, and 0.3, respectively.

Value of X	4	5	6
Probability	0.3	0.4	0.3

(a) Find the mean and standard deviation of this distribution

For a discrete random variable, the mean is found using $\mu = \Sigma x P(x)$.

$\mu = (4)(0.3) + (5)(0.4) + (6)(0.3)$

$\mu = 1.2 + 2.0 + 1.8$

$\mu = 5.0$

Next compute the variance using $\sigma^2 = \Sigma(x - \mu)^2 P(x)$. Since $\mu = 5$, the squared deviations are easy to calculate:

$(4 - 5)^2 = 1, \quad (5 - 5)^2 = 0, \quad (6 - 5)^2 = 1$

$\sigma^2 = (1)(0.3) + (0)(0.4) + (1)(0.3)$

$\sigma^2 = 0.6$

Finally, take the square root to obtain the standard deviation:

$\sigma = \sqrt{0.6}$

$\sigma = 0.774597 \approx 0.775$

Therefore, the mean is 5.0 and the standard deviation is approximately 0.775.

(b) Construct a different probability distribution with the same possible values, the same mean, and a larger standard deviation

To keep the mean equal to 5, it is helpful to choose a distribution that remains symmetric around 5, meaning the probabilities at 4 and 6 are equal. To make the standard deviation larger, we place more probability at the outer values 4 and 6 and less probability at the center value 5.

One valid choice is: $P(X = 4) = 0.4$, $P(X = 5) = 0.2$, $P(X = 6) = 0.4$.

Check that the probabilities sum to 1: $0.4 + 0.2 + 0.4 = 1.0$

Mean: $\mu = (4)(0.4) + (5)(0.2) + (6)(0.4)$

$\mu = 1.6 + 1.0 + 2.4 = 5.0$

Variance: $\sigma^2 = (4 - 5)^2(0.4) + (5 - 5)^2(0.2) + (6 - 5)^2(0.4)$

$\sigma^2 = (1)(0.4) + (0)(0.2) + (1)(0.4) = 0.8$

$\sigma = \sqrt{0.8} = 0.894427 \approx 0.894$

Since 0.894 is greater than the original standard deviation of 0.775, this new distribution has the same mean but a larger spread.

(c) Construct a different probability distribution with the same possible values, the same mean, and a smaller standard deviation

To make the standard deviation smaller, we do the opposite: place more probability at the center value 5 and less probability at the outer values 4 and 6. Again, keeping equal probabilities at 4 and 6 preserves symmetry about 5 and therefore preserves the mean.

One valid choice is: $P(X = 4) = 0.2$, $P(X = 5) = 0.6$, $P(X = 6) = 0.2$.

Check that the probabilities sum to 1: $0.2 + 0.6 + 0.2 = 1.0$

Mean: $\mu = (4)(0.2) + (5)(0.6) + (6)(0.2)$

$\mu = 0.8 + 3.0 + 1.2 = 5.0$

Variance: $\sigma^2 = (4 - 5)^2(0.2) + (5 - 5)^2(0.6) + (6 - 5)^2(0.2)$

$\sigma^2 = (1)(0.2) + (0)(0.6) + (1)(0.2) = 0.4$

$\sigma = \sqrt{0.4} = 0.632456 \approx 0.632$

Since 0.632 is less than the original standard deviation of 0.775, this distribution has the same mean but a smaller spread.

Distribution	P(X=4)	P(X=5)	P(X=6)	Standard deviation
Original	0.3	0.4	0.3	0.775
Larger SD	0.4	0.2	0.4	0.894
Smaller SD	0.2	0.6	0.2	0.632

Interpretation: all three distributions are centered at 5, but they differ in how much probability is placed away from the mean. More probability in the tails produces a larger standard deviation; more probability at the center produces a smaller standard deviation.

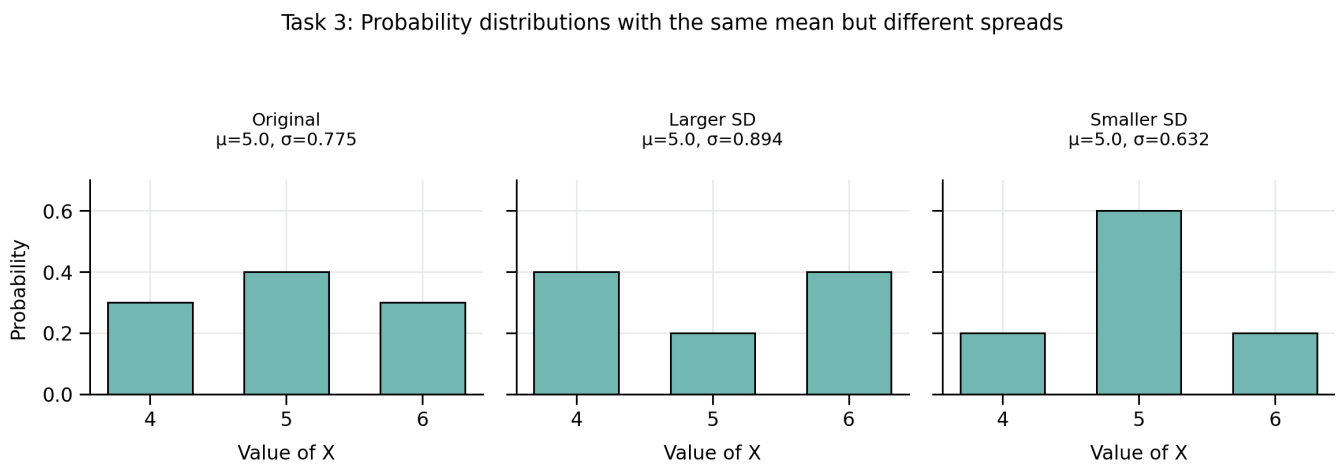

Figure 3. Task 3 comparison of the original probability distribution with alternative distributions having the same mean but different standard deviations.

Task 4 – Salaries and raises (IPS Exercise 2.132)

A hypothetical employee starts working in Year 1 with a salary of $50,000. Each year the salary increases by approximately 5%. By Year 20, the salary is $126,000. The salaries (in thousands of dollars) are shown below.

Year	Salary	Year	Salary	Year	Salary	Year	Salary
1	50	2	53	3	56	4	58
5	61	6	63	7	67	8	70
9	74	10	78	11	81	12	85
13	90	14	93	15	99	16	104
17	109	18	114	19	120	20	126

(a) Describe the relationship between salary and year for this person

A good scatterplot description should discuss the direction, form, strength, and any unusual features.

Direction: the relationship is positive, because salary increases as years of employment increase.

Form: the relationship is approximately linear, since the points follow a straight-line pattern quite closely. However, there is also a slight upward curvature because each annual increase is based on a percentage raise, not a fixed dollar amount.

Strength: the relationship is very strong, because the points are close to the least-squares regression line.

Outliers: there do not appear to be any obvious outliers.

Using the data, the least-squares regression line is:

$\hat{y} = b_0 + b_1x$

$\hat{y} = 41.253 + 3.933x$

Here, x is years of employment and $\hat{y}$ is predicted salary in thousands of dollars. The slope indicates the average increase in salary per additional year according to the linear model.

Slope interpretation: $b_1 = 3.933$ thousand dollars per year

This is about $3,933 per year on average.

(b) The value of r^2 for these data is 0.9832. What percent of the variation in salary is explained by the least-squares linear relationship? Would you say that this is an indication of a strong linear relationship? Explain your answer.

The coefficient of determination r^2 tells us the proportion of the variation in the response variable (salary) that is explained by the least-squares linear relationship with the explanatory variable (year of employment).

Percent explained = $r^2 \times 100\%$

Percent explained = $0.9832 \times 100\%$

Percent explained = 98.32%

Therefore, 98.32% of the variation in salary is explained by the least-squares linear relationship with year of employment.

Yes, this is an indication of a very strong linear relationship, because 98.32% is extremely close to 100%. In practical terms, the line accounts for almost all of the variability in salary. Although the true pattern is not perfectly linear—because a 5% raise each year creates slight curvature—the linear relationship is still very strong and provides an excellent summary of the data.

Additional computational detail (optional): when the data are entered directly, the computed r^2 value is approximately 0.9832, which is consistent with the textbook value 0.9832 after rounding.

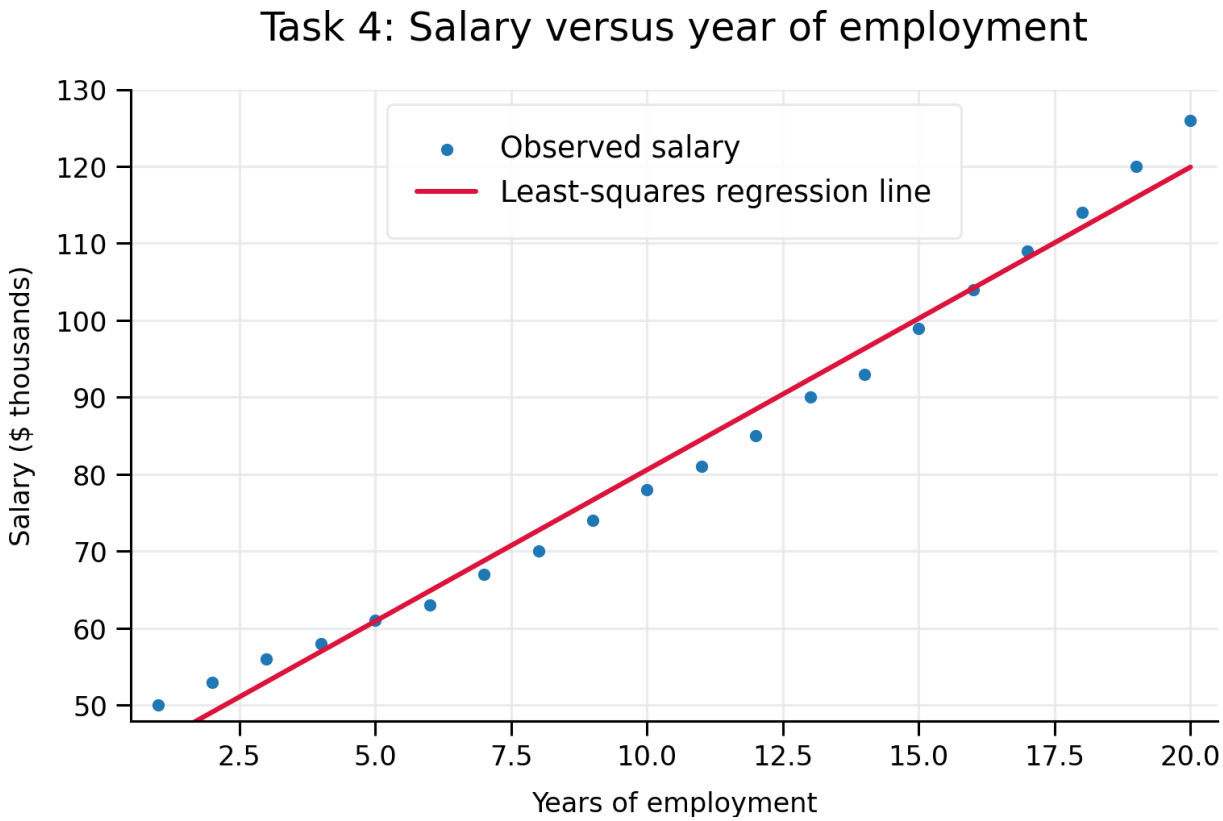

Figure 4. Task 4 scatterplot of salary versus year of employment with the least-squares regression line superimposed.